Case Study: Cooling Time of a Moulded PP Lid

A plant moulds polypropylene (PP) food-container lids in a 4-cavity mould. The lid wall is 2.0 mm thick. A customer asks for a stiffer lid with a 4.0 mm wall. Cooling is the longest part of the moulding cycle, so before quoting a price the plant must know how the cooling time and output change.

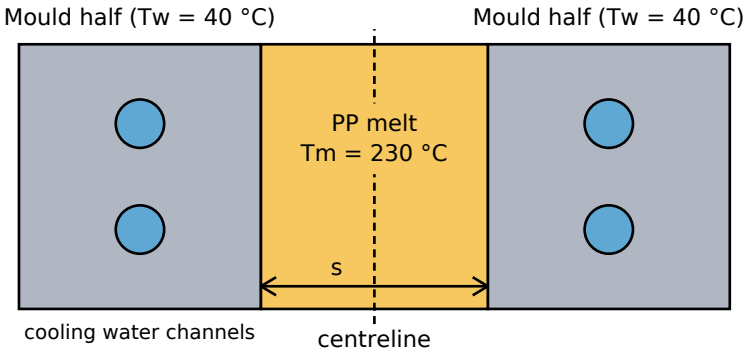

Figure Q1: Cross-section of the lid wall in the closed mould (not to scale).

Given data

Quantity	Symbol	Value
Wall thickness (current / proposed)	s	2.0 mm / 4.0 mm
Melt temperature at end of filling	Tm	230 °C
Mould wall temperature	Tw	40 °C
Ejection temperature (at the wall centreline)	Te	90 °C
Thermal diffusivity of PP	α	0.08 mm²/s
Injection + hold + mould open/eject time	–	6.0 s per cycle (same for both)
Number of cavities	–	4

Assume heat flows only across the wall thickness, both mould faces reach Tw instantly, properties are constant and the latent heat of crystallisation can be neglected. The part can be ejected when its centreline has cooled to Te.

Tasks

1. Set up the transient heat-conduction model for the wall and find the cooling time t_c for the 2.0 mm lid.
2. Plot the centreline temperature against time for the 2.0 mm and 4.0 mm walls (0 to 40 s) and mark the ejection temperature.
3. Find the cooling time, total cycle time and lids per hour for both wall thicknesses.
4. Advise the plant: what does the 4.0 mm wall do to output?